

2A/B PHYSICS
ASSIGNMENT 1: MEASUREMENT

NAME: SOLUTIONS

DUE DATE: _____

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1. What is the difference between a *fundamental unit* and a *derived unit*? Give an example of each.

FUNDAMENTAL: basis of all measurements in science. e.g. m, s, kg, A, K, mol, cd
 DERIVED: combination of fundamental units. e.g. ms^{-1} , ms^{-2} , kgms^{-1} , ...
 (1) (1)

(2)

2. Write the following in *scientific notation*. Include the units.

- (a) 6.43 μm $6.43 \times 10^{-6} \text{ m}$ (b) 293 mm $2.93 \times 10^2 \text{ mm}$
 (c) 0.0085 C $8.5 \times 10^{-3} \text{ C}$ (d) $28.4 \times 10^{-7} \text{ s}$ $2.84 \times 10^{-6} \text{ s}$
 (e) 89.42 nanoseconds $8.942 \times 10^{-8} \text{ s}$
 $8.942 \times 10^{-1} \text{ ns}$ (f) 0.0000412 GJ $4.12 \times 10^4 \text{ J}$
 $4.12 \times 10^{-5} \text{ GJ}$ (6)

3. Convert the following to *standard units*, using scientific notation where possible.

- (a) 0.021 mm $2.1 \times 10^{-5} \text{ m}$ (b) 4953 million kilometres $4.953 \times 10^{12} \text{ m}$
 (c) 26.4 μs $2.64 \times 10^{-5} \text{ s}$ (d) 34.4 m^2 $3.44 \times 10^1 \text{ m}^2$
 (e) $4.85 \times 10^{-3} \text{ mm}^3$ $4.85 \times 10^{-12} \text{ m}^3$ (f) 756 cm^2 $7.56 \times 10^{-2} \text{ m}^2$

(6)

4. How many significant figures are in the following numbers?

- (a) 1.003 4 (b) 0.0021 2 (c) 4.61×10^{-4} 3
 (d) 2×10^3 1 (e) 20000 1 (f) 3000.0 5

(6)

5. Perform the following calculations, giving your answers to the correct number of significant figures.

- (a) $21.6 + 41.24 + 28$ 91 (2)
 (b) $\frac{(61.4)(2.4 \times 10^{-4})}{(3.016 \times 10^{-9})}$ 4.9×10^6 (2)
 (c) $\frac{(28.65 + 7.4 - 3.105)}{(2.649 \times 10^3)}$ $= \frac{33.0}{2.649 \times 10^3} = 1.25 \times 10^{-2}$ (2)

(6)

6. A group of students measured a glass rectangular block with the dimensions 10.52 cm x 20.2 cm x 3.2 cm.

(a) Write the dimensions individually, giving the *absolute uncertainty* and *percentage uncertainty*.

length: $20.2 \pm 0.05 \text{ cm} = 20.2 \text{ cm} \pm 0.25\%$

width: $10.52 \pm 0.005 \text{ cm} = 10.52 \text{ cm} \pm 0.048\%$

height: $3.2 \pm 0.05 \text{ cm} = 3.2 \text{ cm} \pm 1.56\%$

(6)

(b) Calculate the volume of the block, giving your answer in scientific notation with the correct number of significant figures and standard units.

$$\begin{aligned} V &= l \times w \times h \\ &= (0.202)(0.1052)(3.2 \times 10^{-2}) \quad (1) \\ &= 6.8 \times 10^{-4} \text{ m}^3 \pm 1.86\% \quad (1) \\ &= \underline{6.8 \times 10^{-4} \pm 1.3 \times 10^{-5} \text{ m}^3} \quad (1) \end{aligned}$$

(3)

7. A group of students investigated the relationship between the volume of a confined gas and its pressure. They understood that the relationship was given by:

$$P = \frac{k}{V},$$

where P = pressure (cm of Hg)

V = volume (Litres)

k = constant.

Their results are given in the table below.

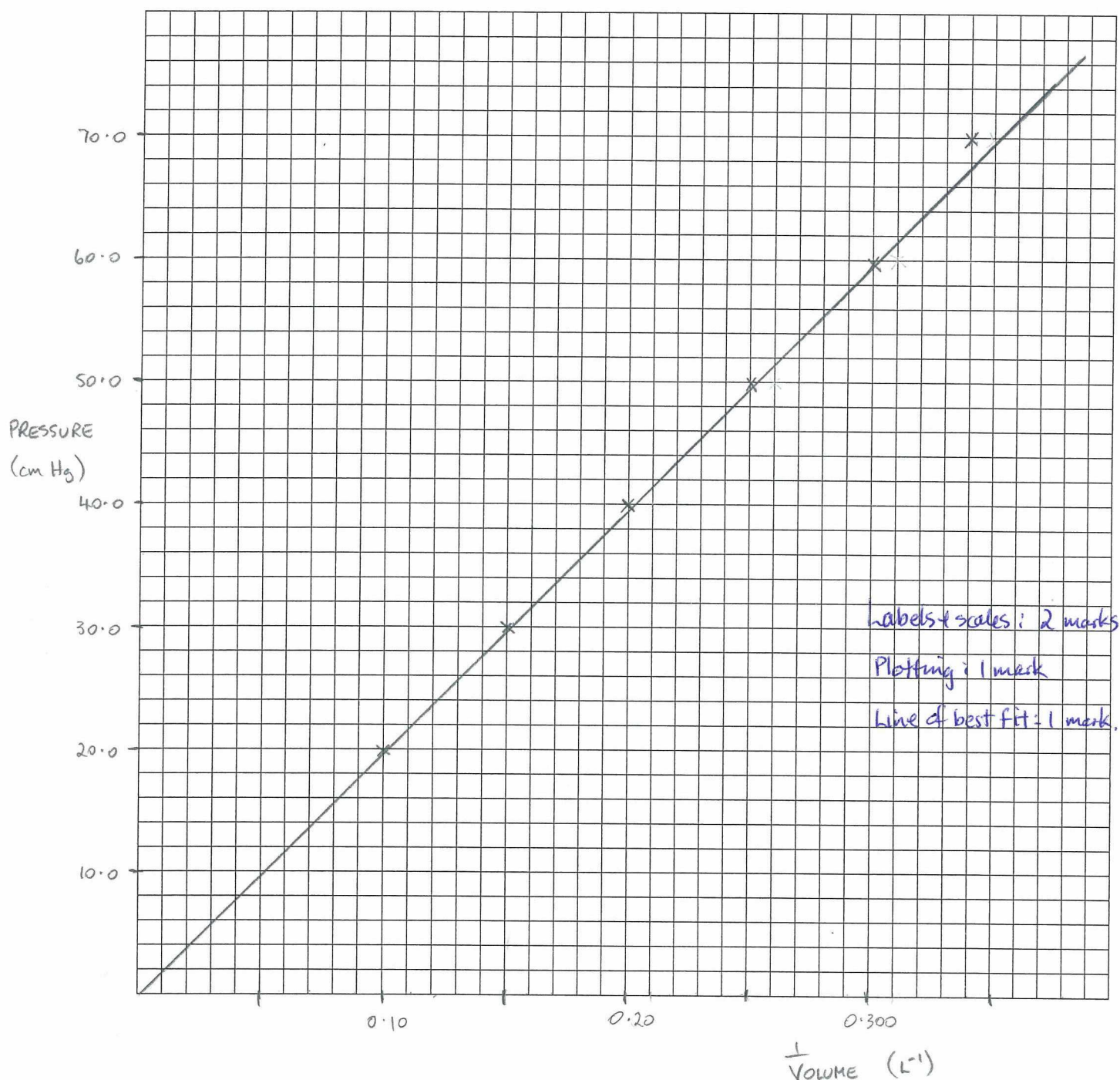
Pressure (cm of Hg)	Volume (L)	$\frac{1}{V} (\text{L}^{-1})$
20.0	10.0	0.100
30.0	6.7	0.15
40.0	5.0	0.20
50.0	4.0	0.25
60.0	3.3	0.30
70.0	2.9	0.34

(a) Manipulate the data to generate a straight-line graph. Record your data in the vacant column.

(2)

- (b) Either graph the data on the grid provided, or use your calculator to generate a line of best fit.

Equation: $P = \frac{1.92 \times 10^2}{V} + 1.22$ (4)



- (c) Use the graph to determine the value of the constant k .

$$\text{gradient} = \frac{(75.0 - 0)}{(0.38 - 0)} = 2.0 \times 10^2 \text{ cmHg L} \quad (2)$$

$$\begin{aligned} P &= \frac{k}{V} \\ \Rightarrow k &= PV \\ &= \text{gradient}_3 \\ &= \underline{2.0 \times 10^2 \text{ cmHg L}} \quad (1) \end{aligned} \quad (3)$$

